

Problem 4

Proving F is a Distribution Function:

We prove the properties required for a cumulative distribution function:

(i) $\lim_{x \rightarrow \infty} F(x) = 1$: Let $x \geq 1$. For all $n \geq 1$, we have $1/n \leq 1 \leq x$, which implies $x \in [1/n, \infty)$. Therefore, $\mathbb{I}_{[1/n, \infty)}(x) = 1$ for all $n \geq 1$. Evaluating $F(x)$ for any $x \geq 1$, we obtain the standard geometric series:

$$F(x) = \sum_{n=1}^{\infty} 2^{-n} = 1$$

Since $F(x) = 1$ for all $x \geq 1$, it follows that:

$$\lim_{x \rightarrow \infty} F(x) = \lim_{x \rightarrow \infty} 1 = 1$$

(ii) $\lim_{x \rightarrow -\infty} F(x) = 0$: Let $x < 0$. For all $n \geq 1$, we have $1/n > 0 > x$, which implies $x \notin [1/n, \infty)$. Therefore, $\mathbb{I}_{[1/n, \infty)}(x) = 0$ for all $n \geq 1$. Evaluating $F(x)$ for any $x < 0$:

$$F(x) = \sum_{n=1}^{\infty} 2^{-n} \mathbb{I}_{[1/n, \infty)}(x) = 0$$

Since $F(x) = 0$ for all $x < 0$, it follows that:

$$\lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} 0 = 0$$

(iii) **Monotonicity** ($x \leq y \implies F(x) \leq F(y)$): Let $x, y \in \mathbb{R}$ such that $x \leq y$, and let $n \geq 1$:

$$\mathbb{I}_{[1/n, \infty)}(x) = 1 \iff x \geq 1/n \implies y \geq 1/n \iff \mathbb{I}_{[1/n, \infty)}(y) = 1$$

Since an indicator function can only take values in $\{0, 1\}$, this implies $\mathbb{I}_{[1/n, \infty)}(x) \leq \mathbb{I}_{[1/n, \infty)}(y), \forall n \geq 1$. Because $2^{-n} > 0$ for all $n \in \mathbb{N}$:

$$\begin{aligned} 2^{-n} \mathbb{I}_{[1/n, \infty)}(x) &\leq 2^{-n} \mathbb{I}_{[1/n, \infty)}(y) \\ \implies \sum_{n=1}^{\infty} 2^{-n} \mathbb{I}_{[1/n, \infty)}(x) &\leq \sum_{n=1}^{\infty} 2^{-n} \mathbb{I}_{[1/n, \infty)}(y) \\ \implies F(x) &\leq F(y) \end{aligned}$$

Thus, F is monotonically increasing.

(iv) **Right-Continuity** ($\lim_{y \downarrow x} F(y) = F(x)$): Let $x \in \mathbb{R}$ and consider a sequence $(y_k)_{k \in \mathbb{N}}$ such that $y_k \downarrow x$. We must show that $\lim_{k \rightarrow \infty} F(y_k) = F(x)$. We have:

$$\lim_{k \rightarrow \infty} F(y_k) = \lim_{k \rightarrow \infty} \sum_{n=1}^{\infty} 2^{-n} \mathbb{I}_{[1/n, \infty)}(y_k)$$

Let $f_k(n) = 2^{-n} \mathbb{I}_{[1/n, \infty)}(y_k)$. Then, $|f_k(n)| \leq 2^{-n} (\forall k \in \mathbb{N}, n \geq 1)$ and $\sum_{n=1}^{\infty} 2^{-n} = 1 < \infty$. Therefore, by the Dominated Convergence Theorem:

$$\lim_{k \rightarrow \infty} F(y_k) = \sum_{n=1}^{\infty} 2^{-n} \left(\lim_{k \rightarrow \infty} \mathbb{I}_{[1/n, \infty)}(y_k) \right)$$

To evaluate $\lim_{k \rightarrow \infty} \mathbb{I}_{[1/n, \infty)}(y_k)$ for $y_k \downarrow x$, we consider two cases:

- If $x \geq 1/n$, then for all k , $y_k > x \geq 1/n$, which implies $\mathbb{I}_{[1/n, \infty)}(y_k) = 1$ for all k . Thus, $\lim_{k \rightarrow \infty} \mathbb{I}_{[1/n, \infty)}(y_k) = 1 = \mathbb{I}_{[1/n, \infty)}(x)$
- If $x < 1/n$, then for sufficiently large k , we have $y_k < 1/n$, which implies $\mathbb{I}_{[1/n, \infty)}(y_k) = 0$. Thus, $\lim_{k \rightarrow \infty} \mathbb{I}_{[1/n, \infty)}(y_k) = 0 = \mathbb{I}_{[1/n, \infty)}(x)$.

Therefore, for all $x \in \mathbb{R}$, $\lim_{k \rightarrow \infty} \mathbb{I}_{[1/n, \infty)}(y_k) = \mathbb{I}_{[1/n, \infty)}(x)$, and:

$$\lim_{k \rightarrow \infty} F(y_k) = \sum_{n=1}^{\infty} 2^{-n} \mathbb{I}_{[1/n, \infty)}(x) = F(x)$$

Because $F(x)$ satisfies the four properties of a distribution function, Theorem 1.4.5 guarantees that F is the distribution function of a unique probability measure $\mathbb{P}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. □

Evaluating Probabilities

The corresponding probability space is modeled by $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{B}(\mathbb{R})$, and probability measure $\mathbb{P}$. $F(x)$ is a step function with jumps only at the points $x = 1/n$ for $n \in \mathbb{N}$. By Lemma 1.4.2, the probability measure $\mathbb{P}$ assigns mass to these specific points equal to the size of the jump:

$$\mathbb{P}(\{1/n\}) = F(1/n) - F(1/n-) = 2^{-n}$$

Thus, $\mathbb{P}$ is a discrete probability measure concentrated entirely on the following set:

$$A = \{1/n : n \in \mathbb{N}\}$$

And the probability of any event $B \subset \mathbb{R}$ is the sum of point masses contained within B :

$$\mathbb{P}(B) = \sum_{x \in A \cap B} \mathbb{P}(\{x\})$$

(i) $\mathbb{P}([1/10, \infty))$: The condition $1/n \geq 1/10$ is satisfied for $n \leq 10$:

$$\mathbb{P}([1/10, \infty)) = \sum_{n=1}^{10} \mathbb{P}(\{1/n\}) = \sum_{n=1}^{10} 2^{-n} = 1 - \left(\frac{1}{2}\right)^{10} = \frac{1023}{1024}$$

(ii) $\mathbb{P}(\{0\})$: The point $x = 0$ is not in the set A , because there is no integer $n \geq 1$ such that $1/n = 0$. Therefore, $\mathbb{P}(\{0\}) = 0$.

(iii) $\mathbb{P}((-5, 1/2))$: The condition $-5 < 1/n < 1/2$ implies $n > 2$ ($-5 < 1/n$ is always true for $n \geq 1$ and $1/n < 1/2$ implies $n > 2$). Since n must be an integer, this means $n \geq 3$:

$$\mathbb{P}((-5, 1/2)) = \sum_{n=3}^{\infty} 2^{-n} = \frac{2^{-3}}{1 - 1/2} = \frac{1/8}{1/2} = \frac{1}{4}$$

(iv) $\mathbb{P}(\mathbb{Q})$: The set $A = \{1/n : n \in \mathbb{N}\}$ consists entirely of fractions, i.e., $A \subset \mathbb{Q}$. Therefore, the points in $\mathbb{Q}$ captures all possible probability mass:

$$\mathbb{P}(\mathbb{Q}) = \sum_{n=1}^{\infty} 2^{-n} = 1$$